

Option D: γ particles (or rays) are uncharged electromagnetic radiation.

30. Technetium-99 has a half life of 6.0 hrs. It is used as a biological tracer. A sample of technetium-99 with an activity of 8.0×10^{10} Bq is injected into the bloodstream of a patient. 20 hours later, when the technetium-99 is assumed to be uniformly distributed throughout the blood of the patient, a 10 cm^3 blood sample is obtained. The activity from the sample is found to be 1.6×10^7 Bq.

What is the total volume of blood in the patient?

- A** 3100 cm^3 **B** 4700 cm^3 **C** 5000 cm^3 **D** 6300 cm^3

Ans: C

First, find activity of the entire sample after 20 hrs.

$$\begin{aligned} A &= A_0 e^{-\lambda t} \\ &= 8.0 \times 10^{10} e^{-(\ln 2 \times 20 / 6.0)} \\ &= 7.937 \times 10^9 \text{ Bq} \end{aligned}$$

$$\begin{aligned} \text{So } \frac{\text{Total volume of blood}}{\text{Volume of blood sample}} &= \frac{\text{Total activity after 20 hrs}}{\text{Activity of blood sample}} \\ \frac{\text{Total volume}}{10} &= \frac{7.937 \times 10^9}{1.6 \times 10^7} \\ &= 5000 \text{ cm}^3 \end{aligned}$$

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